

Rushi Ashok Savani

AI/ML Engineer | Generative AI | LLM Systems | Agentic AI | MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying Generative AI, Agentic AI, NLP, and production ML solutions across healthcare, enterprise, and retail environments. Expertise in LLMs, RAG, text-to-SQL, LangGraph, PyTorch, AWS SageMaker, MLflow, Docker, Kubernetes, and LLMops, with a strong focus on scalable and reliable AI systems. Delivered measurable business impact including 98% answer faithfulness, 90% faster data lookups, 24% fewer AI production issues, and \$1M+ in annual savings.

TECHNICAL SKILLS

Languages & Development: Python, SQL, TypeScript, Node.js, Bash, Git, REST APIs, FastAPI, Postman, Jupyter
Machine Learning: Scikit-learn, XGBoost, Gradient Boosting, SVM, Decision Trees, HDBSCAN, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Classification, Regression, A/B Testing, Hypothesis Testing
Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face Transformers, BERT, RoBERTa, Sentence-BERT, Embeddings, Fine-Tuning, Sentiment Analysis
Generative AI & LLM: LLMs, RAG, Agentic AI, Text-to-SQL, Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, Google ADK, LiteLLM, Prompt Engineering, Guardrails, RAGAS, LangSmith, LLMops
MLOps & Cloud: AWS, Microsoft Azure, SageMaker, EKS, OpenSearch, DynamoDB, Redshift, CloudWatch, MLflow, Docker, Kubernetes, GitHub Actions, CI/CD, Redis, OpenTelemetry, Model Monitoring
Data & BI: Apache Spark, Pandas, NumPy, SQL Server, Snowflake, pgvector, Qdrant, ETL/ELT, Data Modeling, Power BI, Tableau, Streamlit, Excel (Power Query, Power Pivot)

PROFESSIONAL EXPERIENCE

Associate Researcher | Syracuse University Sep 2026 – Present

- Conduct **data analysis** and **research** using Python, SQL, and statistical techniques to analyze complex datasets, identify meaningful trends, and generate actionable insights to support academic research initiatives.
- Run **Hypothesis testing** to identify significant variables for target outcomes, plan feature selection, model choice and validation steps for training and evaluating ML models.
- Collaborate with faculty and research teams to design experiments, document findings, and present research outcomes.

AI Engineer / Data Science | ProHealth Home Health and Hospice | Atlanta, GA Jun 2025 – May 2026

- Built an **agentic text-to-SQL** pipeline on AWS using Python, **LangGraph**, **MCP** tooling, document **knowledge retrieval**, DynamoDB for session memory and self-correcting loops, cutting administrator lookup time by 90%.
- Diagnosed inconsistent retrieval on a 100+ question eval set using **RAGAS** and human-in-the-loop review, fixing with OpenSearch **hybrid search** and reranking, reaching 93% context precision and 98% faithfulness.
- Engineered an **ETL pipeline** using Python, **Apache Spark** and **SQL** to ingest, transform and consolidate patient, financial and operational data into 30+ reusable views, replacing manual processing time by 45%.
- Retrained a readmission ML model on **AWS SageMaker** using **Python**, **scikit-learn** and clinical feature engineering, running 30+ experiments through **MLflow** and cutting reamission cases by 10% (~\$15K savings).
- Architected three **LangChain** pipelines with parameterized SQL and deterministic **chunking** across predefined data points, giving finance and admin teams reproducible, auditable root-cause reports using **LLMs**, saving **\$80K** annually.
- Reduced AI production issues 24% by tracking **token usage**, cost-per query, and hallucination rates live in OpenTelemetry and **LangSmith**, backed by automated evals and **HIPAA-compliant guardrails**.
- Deployed AI Inference services on AWS using **Docker** and **Kubernetes**, automating releases with GitHub Actions **CI/CD** and cutting model-serving costs by 17% via **Redis** caching and routing optimization.

Data Science Research Assistant | Syracuse University | Syracuse, NY Jan 2025 – May 2025

- Fine-tuned a RoBERTa-based sentiment classification model using **PyTorch** and **Hugging Face** Transformers on 7,000+ survey responses, applying class-weight optimization and stratified k-fold validation to improve labeling accuracy by 30%.
- Applied Sentence-BERT **embeddings** and HDBSCAN **clustering** to identify distinct AI-adoption behavior groups from survey data.
- Analyzed sentiment, AI-adoption patterns, and KPIs to derive actionable research insights.
- Developed **Power BI visualizations** and analytical summaries to communicate sentiment distributions, behavioral segments, and **KPI** trends to research stakeholders.

Data Scientist | IBM | Mumbai, India Jan 2023 – Jun 2024

- Mitigated **\$1.2M** in operating expense by lifting baseline precision by 18% with tree-based (XGBoost, Random Forest) and logistic regression models, on **Azure ML** for risk scoring and demand forecasting.
- Improved downstream data ingestion latency 35% by automating ETL and feature engineering pipelines across **Azure Data Factory** and SQL for **1M+** daily transaction records.
- Optimized reporting data pipeline using **star schema** and refactored stored procedures on **Azure PostgreSQL**, bringing query time down 40% and enabling sub-second dashboards for multiple clients.
- Spearheaded rigorous **statistical model validation**, improving model performance through **Bayesian hyperparameter** tuning using **Optuna**, eliminating data leakage and accelerating model selection cycle by 20%.
- Scoped loosely defined business questions from product managers and BI teams into defined model outputs and KPIs, presenting through **Tableau dashboards**, eliminating ad hoc reporting work by 30%.

PROJECTS

PrivateRAG | Python, Qdrant, Llama 3.1, TypeScript, LlamaIndex

Designed a fully local **RAG** system for private-document Q&A with Qdrant vector indexing, metadata filtering, and Llama 3.1 inference, featuring source attribution and chunk-level traceability. | [GitHub](#)

AI Email Response System | Python, RAG, MCP, LLM-as-a-Judge

Implemented an automated RAG-based email response workflow with policy checks, MCP-based routing, **LLM-as-a-Judge** validation, and human handoff for high-risk cases, reducing manual investigation by 70%. | [GitHub](#)

EDUCATION

M.S. Information Systems (Data Science), Syracuse University — GPA: 3.86/4.00 Aug 2024 – May 2026

B.E. Information Technology, Thadomal Shahani Engineering College, Mumbai — GPA: 8.52/10 Aug 2020 – Jun 2024

CERTIFICATIONS

- AI Fluency** — Anthropic, 2026
- AWS Certified Machine Learning Engineer – Associate** — Amazon Web Services
- Generative AI with Large Language Models** — DeepLearning.AI / AWS